

Matihus Molina Larios

Lima, Perú

✉ matihus.molina@unmsm.edu.pe •  maticus •  Maticus-7

Professional Profile

I am a final-year mathematics student at National University of San Marcos with interest in Ergodic Theory, Mathematical applications to health, and Formally verified mathematics. Recently, I have developed a particular interest in the formalization of mathematics using LEAN4. I am also involved in the preparation of research papers related to machine learning applied to lung cancer and a comparative study of numerical methods for differential equations of Lane-Emden type. My current objective is to strengthen my training in pure and applied mathematics and contribute to the development of the academic community in Peru. After graduating, I aspire to pursue postgraduate studies abroad exploring different areas of mathematics.

Education

Universidad Nacional Mayor de San Marcos

Lima, Perú

B.Sc. Student in Mathematical Sciences

03/2022 – 12/2026 (expected)

Cumulative (CGPA): 15.19/20

Cultural Center LIDEMPERÚ

Lima, Perú

Graduate, English Program

2018 – 2021

Languages

Spanish: Native

English: Intermediate B1

Portuguese: Basic A2

Academic Projects

PICV-UNMSM

Project: "A First Encounter with Chaotic Dynamical Systems"

01/2025 – 03/2025

- Academic advisor: Dr. Jorge L. Crisóstomo
- Study of topological conjugacy between Q_c and the Bernoulli Shift.
- Application of concepts from differential equation, functional analysis, topology, and geometry.
- [Click here](#)

Course: Mathematics for ML and DS (UNMSM)

Depression Prediction in University Students

02/2025 – 03/2025

- Application of probability, statistics, and ML techniques.
- Supervised modeling using Python and scikit-learn.
- [Click here](#)

Stellar Structure Modeling

Numerical solution for differential equations of Lane-Emden type

10/2025 – 12/2025

- Analysis of decomposition methods and their differences.
- Numerical implementation and evaluation of the efficiency of these methods.
- [Click here](#)

Academic Events

E'(UBB) = 2025

UBB- Concepción

October 2025

My first international academic experience. I greatly enjoyed the city and the opportunity to learn about a different culture. This experience motivated my interest in pursuing a master's degree abroad.

VII EICM

UNASAM- Huaráz

December 2025

This was my first experience presenting a research project at an event of this scale. The project presented focused on stellar structure modeling via Lane–Emden type equations.

Summer School and Workshop on Optimization and Operator Learning

UNT – Trujillo

February 2026

I participated in this important event organized by SaC³, putting my knowledge of optimization theory and programming to the test, excelling in pre-school.

Research Interests

Pure Mathematics:

- Functional analysis.
- Ergodic theory.
- Mathematical foundations of artificial intelligence.
- Formally Verified Mathematics.
- Mathematical applications to human health.

Technologies:

- Programming languages: Python, R, MATLAB, LEAN4.
- Libraries: NumPy, SciPy, Pandas, Scikit-Learn, Matplotlib, PyTorch.
- Tools: Git, L^AT_EX.

Grants, Distinctions

BECA PERMANENCIA 2025 ,PRONABEC

Recognition from my faculty for my ACADEMIC EXCELLENCE

Extracurricular Activities

Co-Director of the IEEE CIS UNMSM Project and Investigation area

Former member of the cultural and artistic salsa group SALSUNI UNMSM.

Former member of the university judo team UNMSM

Research Trainee of Bildheit

Certificates

Certificates of coursework, academic achievements are available in the following repository.